Lecture 12: Evaluation – Eras, the Anatomy of a Score & Trusting the Numbers

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Course on RLHF and post-training. Chapter 16 + Appendix C.

“Evals are dead, just measure vibes”

This lecture

Vibes are real, but every training decision in this course – data mixes, hyperparameters, which checkpoint ships – is made off benchmark numbers.

Today: where those numbers come from, and when to trust them.

The plan

1. **The eras** – what we measure keeps changing with what we train
2. **Anatomy of a score** – everything between the prompt and the number
3. **Trusting the number** – variance, contamination, and gaming

Part 1: The eras of post-training evaluation

Benchmarks mirror the training goals of their era

The key to reading evals: popular benchmarks are a **reflection of the training best practices of their moment**. When training moves, evaluation follows.

- **Chat era** *(2022-23)* – can it converse like GPT-4?
- **Multi-skill era** *(2023-24)* – post-training improves many capabilities at once
- **Reasoning & tools era** *(2024-26)* – hard problems, long chains of thought
- **Agents & real work** *(now)* – end-to-end tasks inside products and harnesses

Chat era: how close to GPT-4?

Early RLHF models were measured on **chat quality relative to a known strong model**.

- MT-Bench, AlpacaEval, Arena-Hard – and the community-scale version, Chatbot Arena (Chiang et al., 2024)
- The trick that made it scale: **LLM-as-a-judge** replaced human raters (recall Lecture 7 – same machinery as synthetic preference data)
- Narrow by design: these are now just the “chat” and “instruction following” slices of bigger suites

Dubois et al., 2024; Li et al., 2025; Zheng et al., 2023

Multi-skill era: one suite, many capabilities

Once post-training was more than safety and chat, suites like Tülu’s covered:

- **Knowledge**: MMLU, PopQA, TruthfulQA
- **Reasoning**: BigBenchHard, DROP
- **Math & code**: MATH, GSM8K, HumanEval
- **Instruction following & safety** composites

Questions came from the internet; annotation from undergrads and crowdworkers. The flavor:

Example question (MMLU)

What was GDP per capita in the United States in 1850 when adjusting for inflation and PPP in 2011 prices?

(A) About $300 (B) About $3k © About $8k (D) About $15k

Internet trivia more than intelligence – but it tracked pretraining knowledge well.

Hendrycks et al., 2021; Lambert et al., 2024

Reasoning & tools era: make it actually hard

Reasoning models saturated the old suites, so difficulty escalated:

- **Knowledge**: GPQA Diamond, Humanity's Last Exam, FrontierMath
- **Math**: recent AIME contests, run at temperature > 0 with long chains of thought
- **Software**: SWE-Bench (+ variants), LiveCodeBench
- Question sourcing moved from the internet to **grad students, PhDs, and professors** – writing questions became expert labor

Jain et al., 2024; Phan et al., 2025; Rein et al., 2023

Even PhDs and professors are wrong

Example question (Humanity's Last Exam)

What was the rarest noble gas on Earth as a percentage of all terrestrial matter in 2002?

Official answer: **Oganesson**

The official answer is wrong three ways: oganesson is **not a gas** (predicted solid), **not noble** (predicted reactive), and **not terrestrial** (synthetic, ~5 atoms ever made – first synthesized in 2002).

Past a certain difficulty, **verifying the answer key is the bottleneck** – expert-written no longer means correct.

Phan et al., 2025

Today: evals of real work

- The frontier evals are **end-to-end professional tasks**: SWE-bench Verified, Terminal-Bench, GDPVal, APEX
- Task authors are now **experienced professionals**: GDPVal tasks come from experts averaging **14 years** of industry experience; APEX experts average 7+ years at firms like Goldman and McKinsey – expert task-writing is the new cost center
- And the models aren't evaluated bare: they run **inside harnesses and products** (Claude Code, Codex CLI) – last lecture's subject is now the measurement instrument

OpenAI, 2024

Every era ends the same way: saturation

Benchmarks are consumable. As scores approach the ceiling, only the hardest (and mislabeled) items remain, and the benchmark stops separating models.

Part 2: Anatomy of a score

The early-era pipeline was simple

Prompt in, completion out, grade it. Almost everything that could go wrong lived in two places: **how you formatted the prompt** and **how you graded the answer**.

Grading: log-likelihood vs. exact match

Log-likelihood scoring: compare the probability of each answer letter given the prompt. No sampling, deterministic, standard for pretraining evals.

Generation + exact match: sample a completion, regex out the answer. Mirrors real usage – and is standard for post-training.

The catch: rigid format requirements. A model that answers correctly in the *wrong format* scores zero – formatting mismatches can take a model from 60% to near 0.

Brown et al., 2020; Olmo et al., 2025

Prompts evolved with the models

Few-shot prompting → "think step by step" → reasoning models that need no nudge at all.

Modern suites carry **per-benchmark prompts** tuned so formatting isn't the bottleneck – this took real effort to get right.

Tülu 3 MMLU prompt (excerpt)

Answer the following multiple-choice question by giving the correct answer letter in parentheses. Provide CONCISE reasoning for the answer, and make sure to finish the response with "Therefore, the answer is (ANSWER_LETTER)" …

Kojima et al., 2022; Lambert et al., 2024; Wei et al., 2022

Sampling parameters are part of the eval

- Reasoning models need **temperature > 0** for their best scores – Qwen's model cards literally say "**DO NOT use greedy decoding**" (degradation, endless repetition)
- Evaluating at the wrong temperature silently costs points: worth ~3.5 points avg@4 on one benchmark in a recent Qwen3.5 measurement *(per Florian Brand)*
- Read the `generation_config.json` – the recommended settings are "free" performance, and mismatched settings are a silent source of disagreement between reported scores

Then an engine appeared between you and the model

Nobody evaluates raw weights: there is an inference engine or an API in the loop – with its own chat template, tool-call parser, and bugs. vLLM's own postmortem on serving Kimi K2: three engine bugs held tool-call success **below 20%** (218 of 1,200+ calls); after fixes, **99.9%**. Same weights. **APIs do not guarantee correctness either.**

The agentic pipeline: the model is one box of eight

Agentic evals wrap the model in a **harness** (tools, prompts, settings), run it in a **sandbox**, on **hardware**, against tight **timeouts** – and grade hours-long trajectories with regex or an LLM judge.

The harness can be worth months of model progress

- Harness = the tools, prompts, and settings around the model – **the still-unstandardized model side** from last lecture
- Frontier models are **trained in their own harness** – evaluating them in a different one under-reports capability
- The gap is enormous: for Kimi K2, a bad harness cost roughly **6-9 months of model progress** *(per Florian Brand)*
- The extreme case, from the ARC-AGI-3 report: on one environment, Opus 4.6 scores **0% with no harness and 97.1%** with a hand-crafted one
- This is why "same model, different agent product" produces wildly different scores

Hardware is in your score too

- Some benchmarks measure hardware **on purpose**: KernelBench-style tasks need specific GPUs
- Others measure it **by accident**: one resource-hungry command can kill the sandbox and zero the task
- Tight timeouts convert compute into score: reruns of Terminal-Bench 2 with 3-5× timeouts move GPT-5.2 scores by **+6 to +15 points** (community rerun; Brand’s talk) – faster sandboxes mean more iterations before the clock runs out

A score is a property of the system, not the model

Every box below is a knob someone chose – most are undocumented in a press release. Two labs can run “the same benchmark” and measure meaningfully different things.

Part 3: Can you trust the number?

Evaluation variance is everywhere

Sampling at temperature > 0 means **re-running the same eval on the same model moves the score**.

During Olmo 3 we measured it: std. dev. across 3 runs of 14 models, per benchmark →

Most reasoning-era evals sit between **0.25 and 1.5 points of noise** – before anyone changes a prompt or a sampling setting.

More in Appendix C: evaluation variance.

	Benchmark	**σ**
High variance	GPQA	1.48
	AlpacaEval 3	1.24
	IFEval	0.88
Stable	ZebraLogic	0.56
	AIME 24 (avg@32)	0.54
	HumanEvalPlus	0.46
	BigBenchHard	0.39
Very stable	LiveCodeBench (avg@10)	0.29
	MATH	0.25
	MMLU	0.22

Olmo et al., 2025

Managing the noise

- **avg@k is the rescue**: LiveCodeBench was noisy *and* cheap – rerunning 10× moved it from high-variance to very stable. Works everywhere, but balloons costs
- Variance also leaks in from infrastructure: **batch size, tensor-parallel settings, numerics** of long generations
- Practical rule: a **~1-point gap between two press releases is noise**, not signal

Olmo et al., 2025

Why lab-vs-lab comparisons are unreliable

- Each lab's eval stack is **tuned to its internal needs**: custom prompts for key benchmarks, undisclosed formats, different engines
- You see the **output of the function, never the inputs**
- Nobody discloses which public benchmarks were **held out vs. hillclimbed** – train/dev/test hygiene is invisible from outside
- Inference-time scaling confounds everything: more tokens buys more score, and token budgets are rarely controlled

What evals are actually for inside labs

- Labs hillclimb on a few prioritized evals and report the public suite at the end
- The real product of a good internal eval is **statistical power**: less noise on the signals used to compare training runs
- Sometimes the “test set” is just good data: MATH and GSM8K train splits are high-quality – if a lab doesn’t track that eval, training on them is a rational choice
- Human A/B testing and Elo stay in the loop for what benchmarks can’t measure (recall Lecture 8)

Contamination

- **Decontamination** = n-gram / substring search between training and test sets
- Tülu 3 found popular open datasets contaminated: UltraFeedback×TruthfulQA, Evol-CodeAlpaca×HumanEval, NuminaMath×MATH (Lambert et al., 2024)
- The subtle tell: RL with **random rewards** improving Qwen benchmarks – only explicable with contamination in the base model; a real confound in early RLVR research
- Response: perturbed benchmark rewrites (same problem, new numbers) to catch models trained on the original

Huang et al., 2025; Shao et al., 2025; Singh et al., 2024

The model games the eval

Agents love shortcuts – Lecture 9's Goodhart, now holding a terminal. NIST and DebugML have documented these in the wild.

Observed techniques:

- Mining git history for the **future commit that fixes the bug** – one open model did this in **24% of its SWE-bench trajectories**
- Dodging URL blocklists via **mirrors, web archives, and package registries**
- **Hardcoding expected test outputs** into the code
- Abusing quirks of the test runner

Defenses:

- Remove access to everything not strictly needed
- A **second, separate sandbox** for verification and test runs
- A second LLM monitoring the first (expensive)

Grading agents is adversarial now – benchmark design inherits all of reward hacking.

Underelicitation: the score is a lower bound

- ARC-AGI 3 **disallows custom harnesses** in official scoring – “future AGI systems will not need task-specific external handholding” – to keep hand-written rules out of the measurement
- Yet elicitation is where scores come from: a *general* harness (stock Codex CLI + `/goal`, minimal prompt) ran **160 hours and 30K actions to 61%** on the public set, state of the art (source)
- Full elicitation is expensive: in-depth runs of a modern agentic suite can cost **>$100K** *(per Florian Brand)*
- We can only make decisions about **measured** capability – “how good are models at offensive cybersecurity?” and “how big is the open-closed gap?” are only answerable with correct elicitation

Takeaways

- Benchmarks mirror the **training goals of their era** – and every era ends in saturation.
- A score is a **property of the whole system**: prompt, sampling, engine, harness, sandbox, hardware, grader. The model is one box.
- Expect **±1 point of pure noise**; treat cross-lab comparisons as directional at best.
- Contamination, gaming, and underelicitation all bend single numbers – for decisions that matter, **run your own evals** and control the system.

The course so far

0. Prerequisites review
1. Overview *(ch. 1-3)*
2. IFT, Reward Models & Rejection Sampling *(ch. 4, 5, 9)*
3. RL: Motivation & Math *(ch. 6)*
4. RL: Implementation & Practice *(ch. 6)*
5. The Rise of Reasoning Models *(ch. 7)*
6. Direct Preference Optimization *(ch. 8)*
7. Synthetic Data & Modern Post-training *(ch. 12)*
8. Preferences & Preference Data *(ch. 10-11)*
9. Over-Optimization & RLHF's Bad Reputation *(ch. 14, app. B)*
10. Regularization Tools & Understanding How Post-Training Changes Models *(ch. 15)*
11. Tool Use, Function Calling & The Road to Agents *(ch. 13)*
12. **Evaluation** *(ch. 16, app. C) – today*
13. **Crafting Model Character & Products** *(ch. 17) – next (tentative)*

Thank you

Questions / discussion

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